

Cybersecurity Risk Assessment

Athena Learning Inc.

A Qualitative Cybersecurity Risk Assessment of a Cloud-Hosted
EdTech Platform

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Note: Athena Learning Inc. is a fictional organization used in a cybersecurity case study from Cisco (Cisco Networking Academy). This assessment was independently developed to demonstrate cybersecurity risk assessment methodology.

Table of Contents

- 1. Overview
- 2. Business Scenario
- 3. Assessment Objectives & Scope
- 4. Methodology
- 5. Executive Summary
- 6. Key Findings
- 7. Network & Asset Overview
- 8. Risk Register
- 9. Risk Matrix
- 10. Recommended Controls
- 11. Conclusion

1. Overview

This report presents a qualitative cybersecurity risk assessment conducted for Athena Learning Inc., a growing digital education company. The objective of the assessment was to identify critical business assets, evaluate cybersecurity risks, determine the potential business impact of relevant threats, and recommend security controls to reduce operational and information security exposure.

Unlike a vulnerability scan or penetration test, this assessment focuses on risk identification, analysis, and control selection from a business and security perspective, consistent with common industry risk assessment frameworks.

2. Business Scenario

Athena Learning Inc. is a rapidly growing digital education company that delivers online courses, certification programs, and virtual learning services to universities, training organizations, and individual learners across multiple regions. The organization operates a cloud-hosted learning platform, processes online payments, stores student records, supports remote employees, and relies on a combination of cloud and on-premises infrastructure.

Following a period of rapid expansion, executive management requested a cybersecurity risk assessment after concerns were raised about:

- Increasing dependence on cloud services
- Public-facing web applications
- Remote work and personal device (BYOD) usage
- Protection of student and payment data
- Third-party integrations

This assessment was conducted from the perspective of a security analyst engaged to evaluate the organization's overall cybersecurity posture.

3. Assessment Objectives & Scope

Objectives

- Identify critical organizational assets
- Analyze potential threats and vulnerabilities
- Evaluate business impact
- Prioritize significant risks
- Recommend practical security controls

Scope

Category	Examples
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Information assets	Student records, payment data, employee records
Software assets	Learning Management System, SQL databases, web portals
Physical assets	Employee laptops, servers, backup devices
Network assets	Corporate network, DNS services, email infrastructure, remote access systems

4. Methodology

A qualitative risk assessment methodology was used, consisting of the following steps:

- Asset identification
- Threat identification
- Vulnerability assessment
- Likelihood evaluation
- Business impact evaluation
- Risk rating
- Control recommendation

Risk ratings were determined using a likelihood x impact approach and classified as **Low**, **Medium**, **High**, or **Critical**.

5. Executive Summary

The assessment identified several areas requiring immediate attention. The most significant risks involved unauthorized access to student records, exposure of payment information, attacks against public-facing web applications, and security weaknesses introduced through remote work and BYOD usage.

The organization possesses valuable information assets and a broad external attack surface, making identity security, network segmentation, secure application practices, and monitoring capabilities particularly important.

6. Key Findings

Student Data Exposure

Student registration records, authentication data, and personal information represent the organization's most sensitive information assets.

Primary risks:

- Unauthorized access
- Identity theft
- Credential compromise
- Data leakage

Payment Processing Risk

Online payment systems introduce financial and regulatory risk.

Primary risks:

- Payment fraud
- Database theft
- Unauthorized transaction manipulation

Web Application Exposure

The Learning Management System and student portals are publicly accessible.

Primary risks:

- SQL injection
- Cross-site scripting
- Session hijacking
- Authentication bypass

Remote Work and BYOD

Employees access corporate systems from personal devices and remote networks.

Primary risks:

- Credential theft
- Malware infection
- Insecure device configurations
- Data leakage

Cloud Security

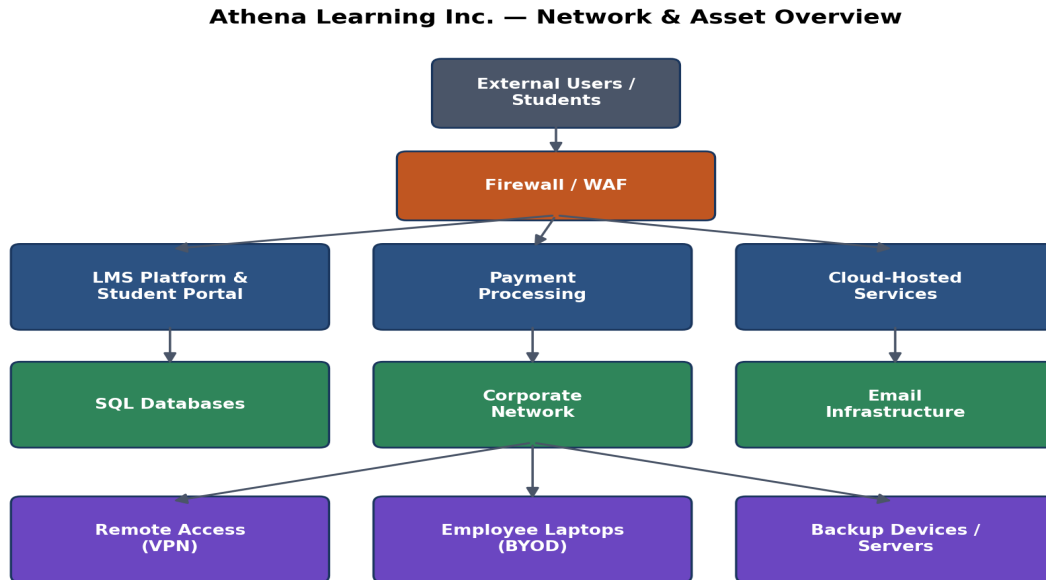
Cloud-hosted services expand operational flexibility but increase configuration risk.

Primary risks:

- Excessive permissions
- Exposed storage
- Insecure identity management
- Monitoring gaps

7. Network & Asset Overview

The diagram below illustrates the high-level relationship between externally facing services, internal systems, and remote/endpoint infrastructure in scope for this assessment.



8. Risk Register

The table below summarizes the key risks identified during the assessment, rated using a likelihood x impact approach. A full version of this register, including recommended controls, is provided as a companion spreadsheet (risk-register.xlsx).

Asset	Threat	Likelihood	Impact	Risk
Student records	Unauthorized access	High	High	Critical
Payment systems	Financial fraud	Medium	High	High
LMS platform	SQL injection	High	High	Critical
Employee laptops	Device theft	Medium	Medium	Medium
Remote access systems	Credential compromise	High	High	Critical
SQL databases	Data corruption	Medium	High	High
Cloud services	Misconfiguration	Medium	High	High
Email infrastructure	Phishing	High	Medium	High

9. Risk Matrix

The risk matrix below was used to translate likelihood and impact ratings into an overall risk classification for each identified risk.

		Risk Matrix		
Impact	High	Medium	High	Critical
	Medium	Low	Medium	High
	Low	Low	Low	Medium
		Low	Medium	High
		Likelihood		

10. Recommended Controls

Identity and Access Management

- Multi-factor authentication
- Role-based access control
- Privileged account monitoring
- Strong password policies

Data Protection

- Encryption at rest
- Encryption in transit
- Secure key management
- Data classification

Network Security

- Network segmentation

- Firewalls
- VPN for remote access
- DNS security monitoring

Application Security

- Secure SDLC
- Input validation
- Parameterized queries
- Web Application Firewall

Endpoint Security

- Full disk encryption
- Endpoint protection
- Mobile device management
- Automated patch management

Monitoring and Detection

- Centralized logging
- SIEM monitoring
- Security alert correlation
- Audit log review

11. Conclusion

The highest priority risks identified in this assessment are concentrated around identity security, public-facing web applications, sensitive data protection, and remote access infrastructure. Implementing the recommended controls, particularly multi-factor authentication, network segmentation, secure application development practices, and centralized monitoring, would meaningfully reduce Athena Learning Inc.'s overall cybersecurity risk exposure.

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